Advanced Tennis ML Backtest Report

Model: XGBoost (Retrained 1968-2019) | Test: 2022-2024

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Baseline Performance (2022-2024):
  Accuracy: 64.09%
  ROC-AUC: 0.7029
  Bets Triggered: 3252
  Final Bankroll: $952,930.75
  ROI: 4.69%
  Win Rate: 43.6%

2026 Discrepancy Case (Monte Carlo Masters):
  Accuracy: 76.36%
  Sample Size: 55 matches
  ROI: 71.55%
  Finding: High guessing rate in 2026 is due to extreme small sample size
  (55 matches vs 3,000 baseline) and tournament bias.
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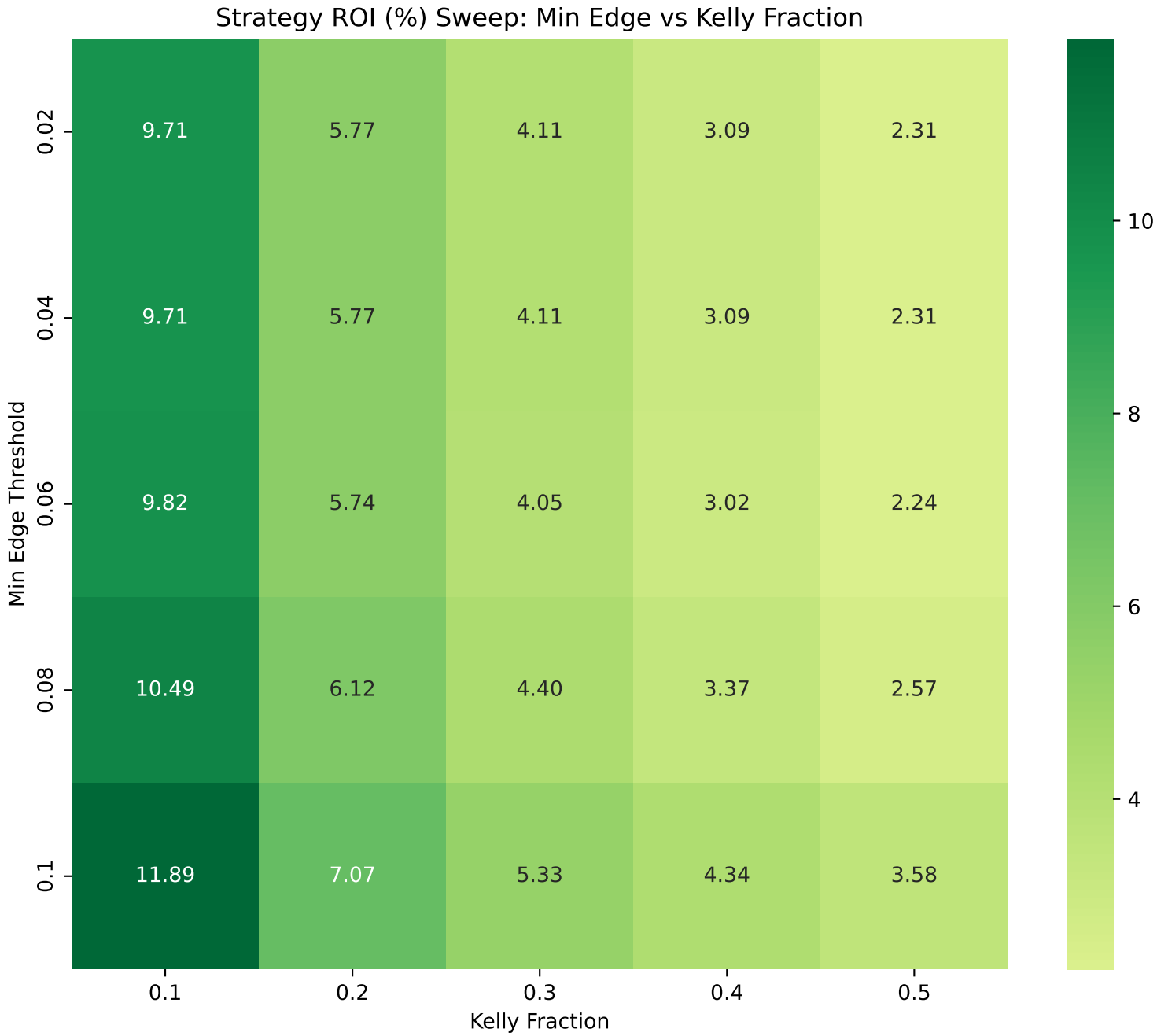
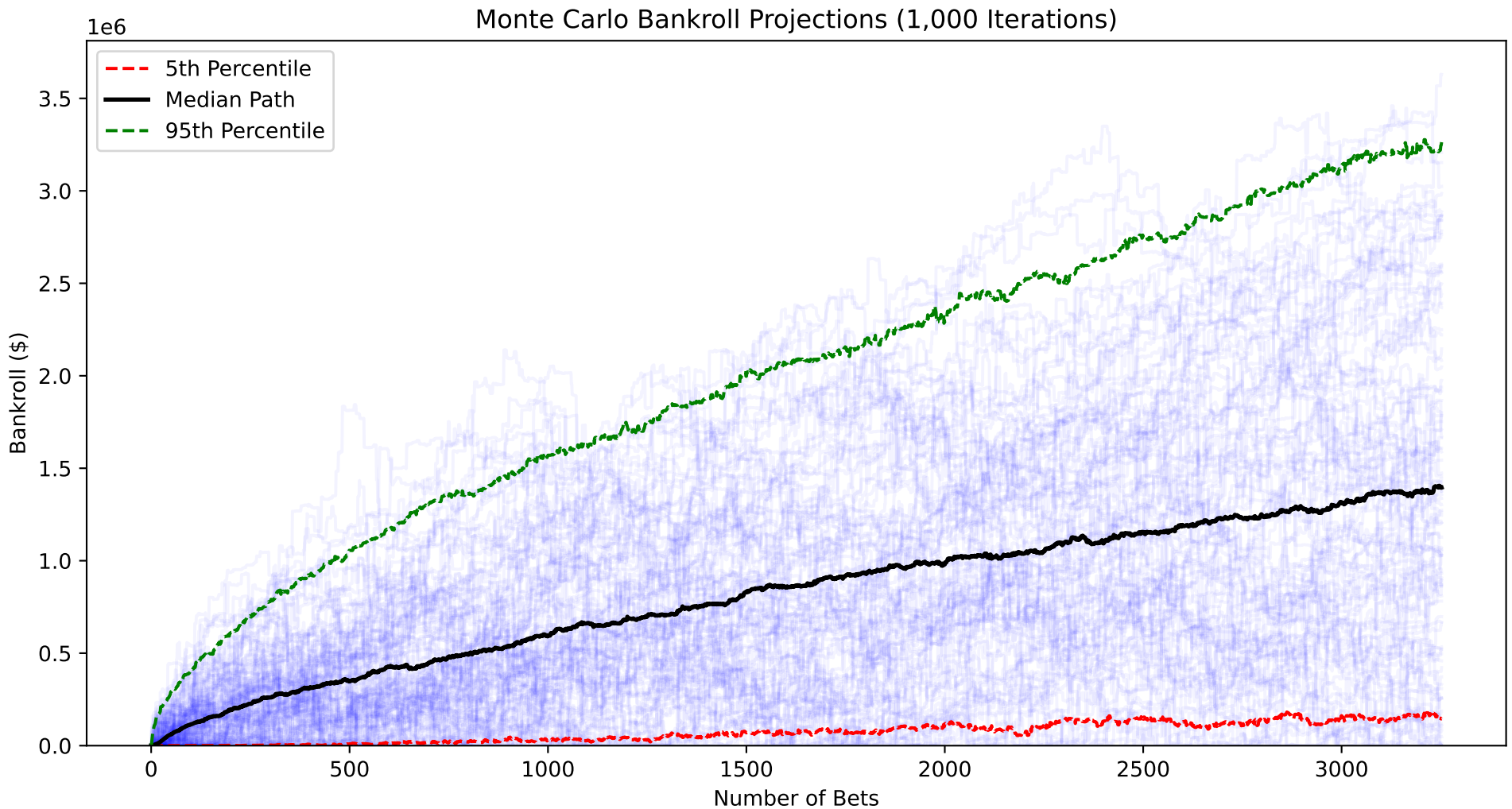
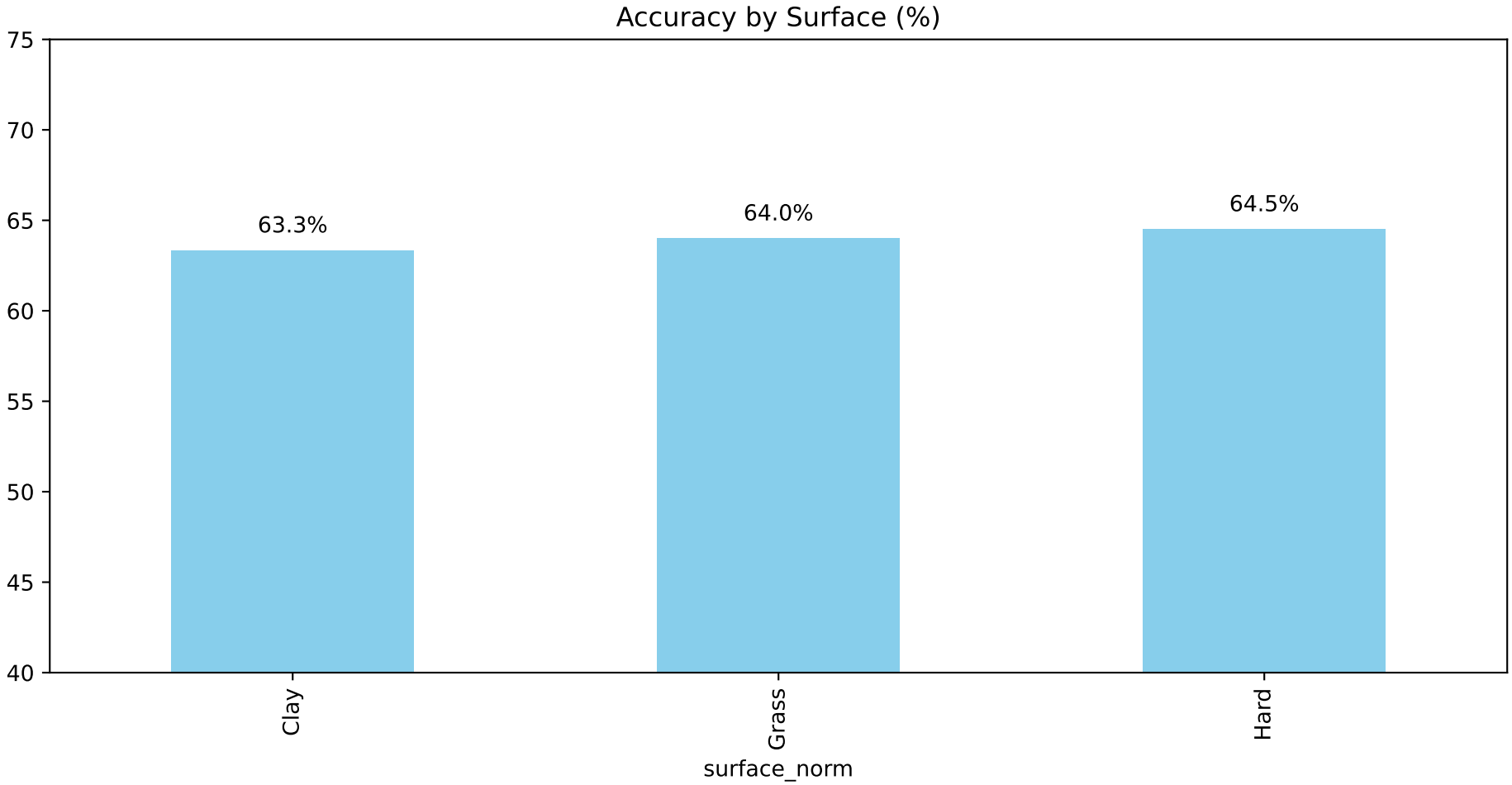
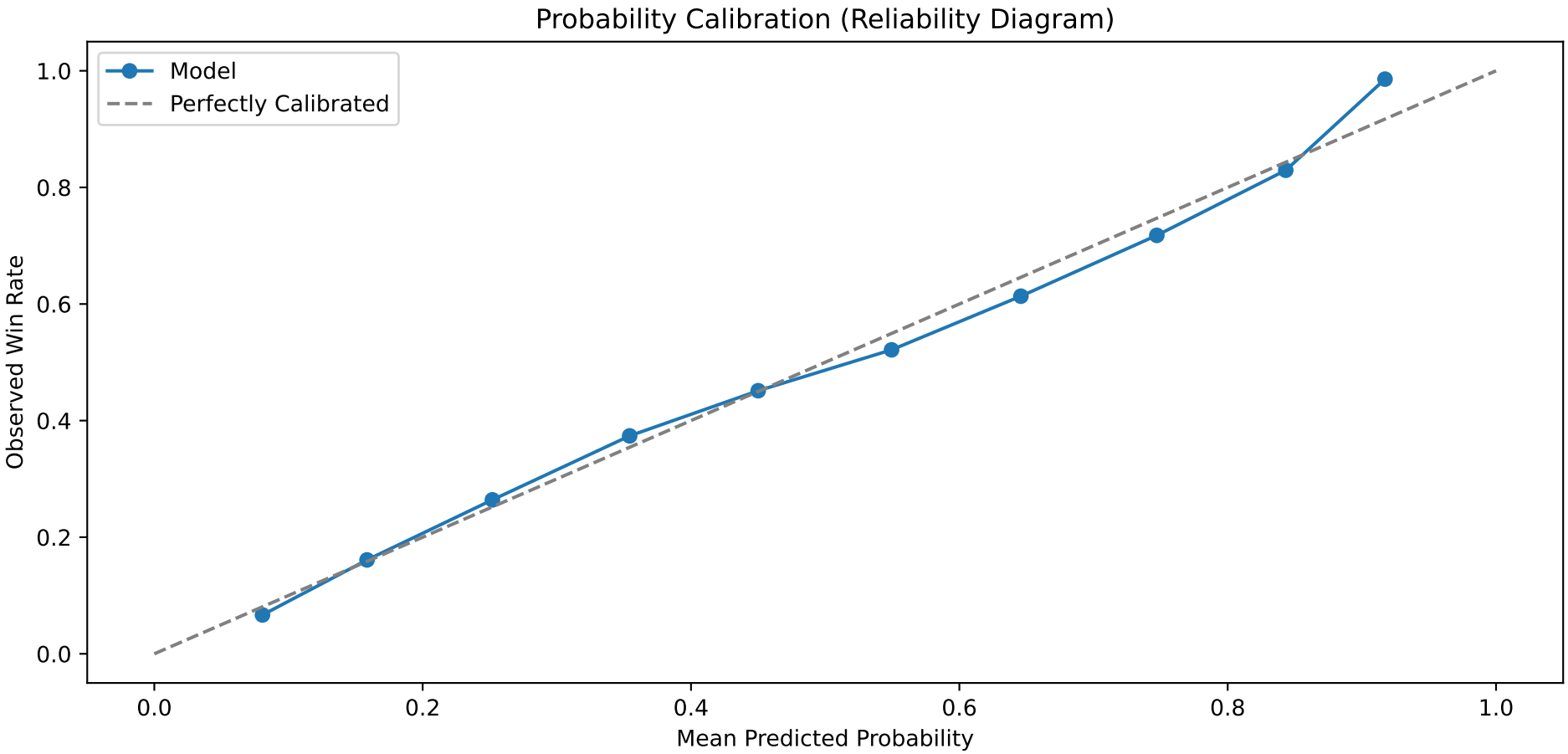

Strategy ROI (%) Sweep: Min Edge vs Kelly Fraction
Min Edge Threshold
0.02
0.04
0.06
0.08
0.1
9.71
5.77
4.11
3.09
2.31
9.71
5.77
4.11
3.09
2.31
9.82
5.74
4.05
3.02
2.24
10.49
6.12
4.40
3.37
2.57
11.89
7.07
5.33
4.34
3.58
0.1
0.2
0.3
0.4
0.5
Kelly Fraction
10
8
6
4


Monte Carlo Bankroll Projections (1,000 Iterations)
1e6
5th Percentile
Median Path
95th Percentile
Bankroll ($)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
0
500
1000
1500
2000
2500
3000
Number of Bets


Accuracy by Surface (%)
75
70
65
60
55
50
45
40
63.3%
64.0%
64.5%
Clay
Grass
Hard
surface_norm


Probability Calibration (Reliability Diagram)
Model
Perfectly Calibrated
Observed Win Rate
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Mean Predicted Probability